SHS

AIRs - LM in

Business Mathematics
Week 4: Module 3

GOVERNMENT PROPERTY
NOT FOR SALE

Business Mathematics

Week 4: Module 3

Target

Different kinds of proportion or proportional relationships are solved in distinct ways.

In your previous module, you have learned the different kinds of proportion and how are they applied in real-life situations.

This module will delve more concepts of proportion. It contains comprehensive lessons on how to solve problems involving the different kinds of proportion, and set of activities to help you practice your skills.

After going through this learning material, you are expected to:

1. solve problems involving direct, inverse, and partitive proportion (**ABM_BM11RP-If-4**).

Before going on, check how much you know about this topic. Answer the activities on the next page.

Discover

In your previous module, you have learned the different types of proportion, which are as follows:

Types of Proportion

1. **Direct Proportion**
   Two variables (x and y) vary such that as x increases, y also increases, or as x decreases, y also decreases proportionally. Or that is, the ratio $\frac{x}{y}$ is always the same. The same holds true with the ratio $\frac{y}{x}$.
2. **Indirect/Inverse Proportion**
   Two variables (x and y) vary such that as x increases, y decreases, or as x decreases, y increases proportionally. Or that is, the product of x and y is always the same.
3. **Partitive Proportion**
   In this type of proportion, a whole is divided into parts.

Now, let's talk about **Constant of Proportionality**. Another way to prove if a given proportion is true is through constant proportionality k. The value of the constant of proportionality depends on the kind of proportion we have between two quantities.

For direct proportion, or that is, y is directly proportional to x, then we have $y = kx$ where k is the constant of proportionality.

For indirect/inverse proportion, or that is, y is inversely proportional to x, then we have $y = \frac{k}{x}$ where k is the constant of proportionality.

The previous module provided you examples of problems/situations involving each kind of proportion. Now, let's recall those examples and solve them one by one, applying the **cross product rule** to solve for the missing terms, and the **constant of proportionality** in order to prove that we have a proportion.

Previous Examples:

1. Two pots of cactus cost Php 130.00. How much do 5 pots of cactus cost? What type of proportion is this?
   Solution: Let x be the missing term or the cost of 5 pots of cactus. That is, $\frac{2}{130} = \frac{5}{x}$. Applying cross product rule, we have, $2x = 650$. Or that is, $x = 325$. The more pots, the higher the cost. That is, both quantities are increasing. Therefore, this is an example of direct proportion.

To check, let's apply the constant of proportionality. Since we have identified this problem as direct proportion, our equations will be in the form $y = kx$. That is, $2 = k(130)$ and $5 = k(325)$. Or that is, $k = \frac{1}{75}$ and $k = \frac{1}{75}$, respectively. Since we have an equal value of k, we can conclude that this is a proportion.

2. Two men can finish planting rice in a hectare of rice field for four days. At the same rate, how many days will it take for 8 men to do the job? What type of proportion is this?
   Solution: Let x be the missing term or the days it will take for 8 men to finish the job. That is, $2(4) = 8(x)$. Or we have, $x = 1$. The more number of workers, the lesser number of days to finish the job. That is, as one quantity increases, the other quantity decreases. Therefore, this is an example of indirect/inverse proportion.

   To check, let's apply the constant of proportionality. Since we have identified this problem as indirect/inverse proportion, our equations will be in the form $y = \frac{k}{x}$. That is, $2 = \frac{k}{4}$ and $8 = \frac{k}{1}$. Or that is, $k = 8$ and $k = 8$, respectively. Since we have an equal value of k, we can conclude that this is a proportion.

3. A cash incentive worth Php 30,000.00 is to be divided by 3 families with the ratio 1:3:4, how much does each family receives? What type of proportion is this?
   Solution: The ratio 1:3:4 indicates that $1 + 3 + 4 = 8$ portions. Therefore, Php 30,000.00 is to be divided into 8 portions. That is, $\frac{30{,}000}{8} = 3{,}750$. Or that is, Php 3,750.00 per portion.

   $$\text{Family A (1 portion)} = 1(\text{Php } 3{,}750.00) = \text{Php } 3{,}750.00$$
   $$\text{Family B (3 portion)} = 3(\text{Php } 3{,}750.00) = \text{Php } 11{,}250.00$$
   $$\text{Family C (4 portion)} = 4(\text{Php } 3{,}750.00) = \text{Php } 15{,}000.00$$

This shows division of whole into parts. Therefore, this is an example of partitive proportion.

Gauge

Congratulations! You've earned all the badges. Time to use your skills!

Directions: Read and analyze the following problems carefully. Use separate sheet to solve for the values of x. Show complete solution.

1. Jera and his sister can finish painting their whole house for 5 days. At the same rate, how many days will it take them together with their mom and dad to finish the job?
2. A copy machine can duplicate 2,400 copies in one hour. How many copies can it make per minute?
3. Patient XYZ is receiving 1 liter of IV fluids every 8 hours. At that rate, how much will he receive in 3 hours?
4. If it takes two weeks for 4 men to build a house, how many days will it take for 9 men to do the same job? What type of proportion is illustrated? Why do you think so?
5. A liter of lemon juice is to be divided among the three kids, Ciara, Dane, and Nash in the following portion: Dane's share is thrice as Ciara's while Nash's share is four times as Ciara's. How many milliliters of juice will each one get?
6. Covy consumes 23 megabyte (MB) of mobile data in 30 minutes. How many megabytes of mobile data can he consume in 8 hours?
7. If a 2-liter carbonated drink contains 60 kilocalories for every 200 ml, how many kilocalories does it contain in total?
8. If it takes 125 days for 5 members to complete a research, how many members are need to complete the same research in 25 days?
9. Three friends Judy, Shana, and Maria put up a tea shop. They agreed to divide their monthly net income with the ratio 1:3:4 respectively, according to how much they've contributed in the capital. If their net income for this month is Php 8,400.00, how much does each one gets?
10. If 8 men can deliver 500 parcels in one day, how many days will it take for 3 men to deliver the same parcels?
11. In a recent election for SSG President, Inigo received 7 votes for every 5 votes that Joao receives. How many votes did each of them received if 444 votes were cast?
12. Jase spends 7 hours a day using his gadgets. How many hours of using gadgets did he spent for the month of July?
13. In an orchard of 2,700 trees, there are mangoes, *duhat*, and *rambutan* in a ratio of 7:3:2. How many trees of each kind are there?
14. A 2000-watt water heater can boil water in 240 seconds. If the power is reduced to 1000 watts, how long will it take to boil water?
15. The length of three ring light stands are in the ratio 1:2:4. If their total length is 4 meters, how long is each stand?

Business Mathematics

Week 5a: Module 4

Target

A *buy-and-sell* enterprise is nothing more than purchasing cheaply new or previously owned products that we all need, use or want, and reselling these same items for more than cost. The difference between what you paid and what you sold it for, of course, is your *profit* (Stephenson,2005).

Buying ang Selling is said to be the New Economy. Many people choose to buy and sell because it requires low investment and minimal skills, it has minimal financial risk, it is incredible for profit potential, and you can work from home.

In your previous lesson, you learned how to solve problems involving fractions, decimals and percent. Buying and selling is one of the interesting applications of percent in real life.

This module will provide you with information and activities that will help you understand the key concepts in buying and selling.

After going through this module, you are expected to:

1. Differentiate Mark-on, Mark down and Mark-up (ABM_BM11BS-Ig-1

Before going on, check how much you know about the topic.
Answer the pretest on the next page in a separate sheet of paper.

LESSON 1
CONCEPTS ON BUYING AND SELLING

Discover

To understand the concept of Mark-on, mark-up and mark down, let us first define these important terms used in business.

PRICING

Setting prices is another application of our knowledge on percentage. Trading or merchandising firms and manufacturing firms make use of of pricing decisions. Trading or merchandising firms are those who do "buy and sell". It means they sell what they buy. Manufacturing firms are those who buy raw materials, process them and sell finished products.

Setting the right price is important for goods to sell. If the price is too high, the customers may not be able to afford it; if it is too low, the company may not be able to make a profit. Aside from the cost of the product, operating expenses should be considered for the company to earn a profit.

To be able to set the correct selling price of your product, you have to understand the following terms: cost price, operarting cost, selling price, profit, mark-up, mark-on and markdown.

COST PRICE

This refers to the purchase price of a product that a company or store is going to sell. For example, if a store owner bought a T-shirt for PhP150 this is the cost price of the product as far as the store owner is concerned.

It is also the price that has to be spent to produce goods or service before any profit is added. This is usually computed on per unit basis.

In our previous activity, the cost price refers to the price of raw materials needed to produce a *Camote* cue. For instance, if the total cost of the raw materials is PhP400 which will produce 50 sticks of *Camote* cue, divide the total cost with the number of sticks to be sold. Then, the cost price of each stick is PhP8.

OPERATING COST

This refers to the price (per unit) incurred relative to the production and sale of commodity. This includes other expenses like rent, staff salaries, travel expenses and other incidental expenses.

Operating cost can also be set as a percentage of the cost price. For example, the operating cost for a T-shirt purchased from divisoria at PhP200can be set at 25% of the cost price. In this case, the operating cost is PhP50 (multiply rate of operating cost 0.25 with cost price 200).

In our previous activity, the operating cost refers to the other expenses incurred to produce *Camote* cue. For Instance, if the total operating expenses in producing 50 sticks of *Camote* cue is PhP150, then the operating expense per stick is PhP3 (divide total cost by number of sticks produced).

SELLING PRICE

This refers to the price at which the product or item is sold per unit. The selling price is obtained using the given formula below.

SELLING PRICE = COST PRICE + OPERATING EXPENSES+PROFIT

S = C + E + P

Where S = Selling Price
C = Cost Price
E = Operating Expenses
P = Profit

PROFIT

It is the money earned after the cost price and operating expenses are accounted for after the sale of a commodity.

In our previous activity, we assumed that the estimated cost of raw materials is PhP400 and the other expenses is estimated to be PhP 150. From the raw materials, bought we estimated that 50 Camote cue sticks will be sold. With that, cost price is divided by 50 to obtain the cost per stick of camote cue. That is PhP 8 per stick. The operating cost per unit is also obtained by dividing the estimated total expenses by 50. Then the operating cost per stick of camote cue is PhP3. If we set our profit to be 25% of the cost price per stick, we obtain the amount of profit by multiplying .25 with 8 that is PhP2 per stick. We can now obtain the selling price of the camote cue using the given formula.

S = C + E + P
S = 8 + 3 + 2
S = 13

Therefore, for us to achieve the desired profit of 25%, we have to sell a stick of *Camote* cue at PhP13.

After learning how to set the selling price of a product, let us learn what mark up, mark on and mark down is all about.

MARK-UP

This refers the amount added to the cost price to determine the selling price. It is also defined as the difference between the selling price and the cost price and sometimes refered to as GROSS PROFIT. It can also be expressed as a percentage above the cost.

For example, a T-shirt bought at PhP150 is sold at PhP200, the difference which is PhP50 is the mark-up of the T-shirt. In our previous activity, the mark-up is PhP5 because the cost of each Camote cue is PhP8 but sold at PhP13 each.

Mark-up value is usually computed based on cost. Mark-up in percent or mark-up rate on the other hand can be computed based on either cost price or selling price.

MARK-ON

This refers to the additional increase in the price of the product or item most common during various seasons or holidays like two weeks before Cristmas due to the increase of demand. Mark-on is usually computed based on selling price.

For example during the food sale, you realized that there is a high demand for your *Camote* cue and you want to take advantage of this peak season, thus you decide to increase the price from PhP13 to PhP15. The additional increase on the price is called mark-on.

The other reasons why mark-on becomes an option for the business owner are the following:

1. A calamity has hit the source of raw material or commodity therefore affecting its supply. (price of alcohol increased due to pandemic)
2. Seasonal Demands. (Christmas, Valentine's , etc.)
3. Special occassions is being celebrated. (commemorative items like t-shirts, etc.)

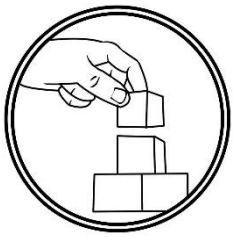
MARK-DOWN

This refers to reduction in the selling price of an item to stimulate its demand or to drive a competitor out of the market. Sometimes, markdowns are created to remove a slow-selling item from the inventtory. The rate of mark-down is always computed based on selling price.

The other reasons why mark-downs are sometimes thought of as an option for a business owner are the following:

1. The item is a perishable item and it is best to dispose of it sooner that simply throw it away. (cooked food, fruits, vegetables, etc.)
2. The item has become dirty or worn out, or possibly out of style.
3. Competition forces the marking down of an item.

After considering that the sale of your Camote cue is coming low, you may need to sell it at a **DISCOUNTED PRICE** or what is commonly known as **SALE PRICE**. You may decide to sell you Camote cue at a discounted price of PhP11 which covers the cost price and the operating expense but zero or no profit. The price which give a zero profit is also called the **BREAK-EVEN PRICE**. You may also decide to simply sell it at PhP8 each to only cover the cost price. With this, price, you will have a negative profit which is said to be **LOSS.**



Explore

Here are some enrichment activities for you to work on to master and strengthen the basic concepts you learned from this lesson

Activity 1.Can you Tell?
Directions: Tell whether the situation is an example of Mark-up, Mark-on or Mark-down.

SITUATIONS	ANSWER
1. A T-Shirt bought at PhP150 was sold for PhP200.	
2. A blouse originally priced at PhP500 is on sale for only PhP350.	
3. A chocolate bar originally priced at Php80 is now being sold for Php100.	
4. The cost of a dozen of apple is PhP200. A store owner sell each apple for PhP20.	
5. Carl bought a pair of shoes regularly priced at Php4,000 for only PhP2,500.	
6. A story book regularly priced at PhP850 was sold at 30% discount.	
7. Ben is waiting for a pair of sunglasses curretly priced at PhP1,200 to be sold at a sale price of PhP800.	
8. A jacket which costs PhP950 is being sold at P1,575.	
9. During All Saint's Day, a bunch of flowers regularly priced at PhP150 was being sold for PhP200.	
10. An on-line seller wants to realize a PhP50 profit and set the operating expenses at 25% based on cost.	

Activity 2. Do you Agree?
Directions: Read the statements below. Decide whether you agree or not. Check the box corresponding to your answer.

Do you agree with these statements?	YES	NO
1. Mark-up is the sum of the operating expense and profit.		
2. Mark-on becomes an option to store owners because of insufficient supply.		
3. Mark-down helps to stimulate sales of slow selling products.		
4. Mark-up is also called Gross Margin.		
5. Mark-on becomes an option to store owners to take advantage of the high demand.		
6. Mark-down is an option for store owners to clear the inventory.		

Target

One of the best aspects about starting and operating a buy-and-sell venture is that everyone is qualified. Buying and selling knows no boundaries-anyone with a need or desire to earn extra money, work from home or to start and own a business can buy and sell, regardless of age, experience, education and financial resources. This is perhaps what makes buying and selling the ultimate self-employment option for the vast majority of people; it's cheap, easy, quick and proven to work and generate huge profits (Stephenson,2005).

Profit is a reward for risk-taken in the business. The main goal of a business is making **profit**.

This module will provide you with information and activities that will help you understand how to obtain mark-up, mar-on and mark-down given the price of the product. This will enable you to keep track of the profit you can gain in every business transaction you make.

After going through this module, you are expected to:

1. Obtain Mark-on, Mark down and Mark-up given price of a product.

Before going on, check how much you know about the topic.
Answer the pretest on the next page in a separate sheet of paper.

LESSON 2
MARK-UP, MARK-ON and MARK-DOWN

Discover

Let us continue to understand how Mark-on, mark-up and mark down are obtained. Remember the important facts you learned previously.

MARK-UP

It is the amount added to the cost price to determine the selling price. It is also defined as the difference between the selling price and the cost price and sometimes refered to as GROSS PROFIT. .

To compute Mark-up: MARK UP = SELLING PRICE – COST PRICE

MU = S – C

Where MU = Mark-up
S = Selling Price
C = Cost Price

Illustrative Examples:

1. Jose buys a T-shirt with a cost of PhP175. If he sells the T-shirt at PhP245,
   a. How much is the mark-up?
   b. What is the rate of mark-up based on cost?

Given: Cost (C)= 175
Selling Price (S) = 245

Solution:

a. Solve for the amount of mark-up using the formula MU= S – C.
   Then, MU = 245 – 175 by substitution
   = 70
   The mark-up is PhP70.

b. Solve for the rate of mark-up by dividing the mark-up by the cost
   Then, $\mathbf{MU_c\% = \frac{MU}{C} \times 100\%}$
   Rate of mark-up = $\frac{70}{175}$ x 100% by substitution
   = 40%

   The T-shirt has a 40% mark-up based on cost.

2. Aling Maria would like to sell face masks purchased for PhP20 each. If the operating cost is set at 25% of the cost and she would like to have a profit of 15% on the cost of each item, help aling Maria determine the selling price for each mask.

Given: Cost (C) = 20
Rate of Expenses = 25% or 0.25
Rate of Profit = 15% or 0.15

Solution:

a. Represent the Operating Expenses (E) and Profit (P) as percentage of cost.
Remember that operating expense and profit are computed based on cost so we have to multiply the given rate of expense and profit to the cost.
Then, E = 0.25 x C and
P = 0.15 x C

b. Solve for the amount of mark-up Using the formula MU = S – C
Since the selling price is not given but we know that S = C + E + P,
We substitute the value of S to MU = **S** - C
then we get MU = **C + E + P** – C
we now have a new formula **MU = E + P**

From MU = E + P
MU = (0.25 x 20) + (0.15 x 20) by substitution
= 5 + 3
= 8
The mark up price for each mask is P8.00.

c. Solve for the selling price using the formula
S = C + E + P but MU = E + P , then
S = C + MU then suctitute the value of C and MU
= 20 + 8
= 28
Aling Maria should sell each face mask at PhP28.

Alternative solution:

S = C + E + P	but E = 25% of the cost or 0.25C
	and P = 15 % of the cost or 0.15C
= C + 0.25C + 0.15C	by substitution,
= 1.40C	by addition of decimals
= 1.40 x 20	by substitition , C=20
= 28	

MARK-ON

It is the difference between the Peak Selling Price and the Regular Selling Price.

To compute Mark – On:

MARK-ON = PEAK SELLING PRICE – REGULAR SELLING PRICE

MO = PS – S

where MO = mark-on
PS = Peak Selling Price
S = Regular Selling Price

Illustrative Examples:

1. The regular price of 500mL bottle of alcohol in a certain store is PhP65. Due the pandemic, there is high demand for alcohol. The store owner then decided to change the price tag to PhP92.50.
   a. What is the amount of mark-on?
   b. What is the rate of mark-on?

Given: Regular Selling Price (S) = 65
Paek Selling Price (PS)= 92.50
Solution:

a. Solve for the amount of mark-on using the formula MO = PS - S.
   MO = 100 - 65 by substitution
   = 35

b. Solve for the rate of mark-on based on selling price using the formula
   $$\mathbf{MO_S\% = \frac{MO}{S} \times 100\%}$$
   Rate of mark-up = $\frac{32.50}{65}$ x 100% by substitution
   = 50%

   The amount of mark-on is PhP32.50 which is equivalent to 50% mark-on.

2. Manang Biday observed that there is an undersupply of fish so the market goers prefer to buy dressed chicken. She then decides to increase the price of dressed chicken by 15% of the regular selling price per kilo. If the cost of dressed chicken per kilo is PhP150 with a 20% mark-up, what is the peak selling price?

Given: Cost (C)= 150
Rate of Mark-up (MU_C%) = 15% or 0.15
Rate of Mark-on (MO_S%) = 20% or 0.20
Solution:

a. Compute for the amount of mark-up which is based on cost. We then multiply the rate of mark-up with the cost price.
   MU = 0.20 x 150
   = 30
b. Compute for the regular selling price by adding the mark up with the cost
   SP = C + MU
   = 150 + 30 by sustitution
   = 180
c. Compute for the amount of mark-on by multiplying the rate of mark-on with the selling price. MO = 0.15 x 180
   MO = 27
d. Finally, compute for the peak selling price by adding the mark-on and selling price. From MO = PS – S, it follows that
   PS = S + MO
   PS = 180 + 27 by sustitution
   = 207

The peak selling price of the dressed chicken is PhP207.

MARK-DOWN

It is the difference between the Selling price and the Sale Price

To compute for Mark-down: MARK-DOWN = SELLING PRICE – SALE PRICE

MD = S – SP

where MD = Mark-down
S = Regular Selling Price
SP = Sale Price

Illustrative examples:

11. Miguel was able to buy a smart cellphone regularly priced at PhP9,990 for only PhP7,492.50.
    a. What is the amount of mark-down?
    b. What is the rate of mark-down?

Given: Regular Selling Price (S) = 9990
Sale Price (SP) = 7492.50

Solution:

a. Solve for amount of mark-down using the formula MD = S –SP
   MD = 9990 – 7492.50 by sustitution
   = 2497.50
b. Solve for the rate of mark down. Note the rate of mark down is based on selling price. Use the formula $\mathbf{MD_S\% = \frac{MD}{S} \times 100\%}$

   $MD = \frac{2497.50}{9990} \times 100\%$ by sustitution
   = 25%

   The amount of mark-down is PhP2497.50 which is equivalent to 25% mark-down based on selling price.

12. A stall owner paid PhP20,000 for a flat television. Expenses are 20% of the selling price while the required profit was 17% of the selling price. During the inventory sale, the TV was marked down by 25%.
    a. What was the selling price?
    b. What was the sale price?

Given: Cost (C) = 20,000
Rate of Expenses = 20% or 0.20
Rate of Profit = 17% or 0.17
Rate of Mark-down (MD_S%) = 25% or 0.25

Solution:

a. Represent the expenses and profit as percentage of cost.
   E = 20% of cost or 0.20C ,
   and P = 17% of cost or 0.17C.
b. Compute for the selling price using the formula S = C + E + P.
   S = C + 0.20C + 0.17C by sustitution
   = 1.37C but C = 20,000
   = 1.37 x 20000

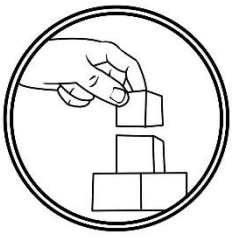
= 27400

The regular selling price of the flat TV is PhP27,400.

c. Compute for the sale price considering that the rate of mark-down is 25% of the regular selling price.

SP = S – MD but MD = is 25% of selling price or 0.25S
= S – 0.25S by substitution
= 0.75S but S = 27,400
= 0.75 x 27400
= 20550

The sale price of the flat TV was PhP20,550.



Explore

Here are some enrichment activities for you to work on to master and strengthen the basic concepts you learned from this lesson

Activity 1. Complete Me!

Instructions: Complete the table by computing the required value using appropriate formula.

A. Mark-Up

Cost (C)	Selling Price (S)	Mark-Up (MU)	Mark-Up rate based on cost (MU_c%)
1. _____	PhP480	PhP120	_____
2. PhP300	PhP405	_____	_____
3. PhP400	_____	_____	30%

B. Mark-On

Original Selling Price (S)	Peak Selling Price (PS)	Mark-On (MO)	Mark-On rate based on Selling Price (MO_S%)
4. PhP280	PhP320	_____	_____
5._____	PhP800	PhP120	_____
6. PhP1,520	_____	_____	15%

C. Mark-Down

Original Selling Price (S)	Sale Price (SP)	Mark-Down (MD)	Mark-Down rate based on Selling Price (MD_S%)
7. PhP415	PhP385	_____	_____
8._____	PhP620	PhP74.40	_____
9. PhP3,250	_____	_____	20%
10._____	_____	PhP150	25%

Activity 2. Identifying Given Values
Instructions: Identify the the given values for each situation below.

A. I am waiting for a pair of sunglasses currenttly priced at PhP 1,200 to be sold at a sale price of P850. What rate of mark down will be necessary for this to happen?
   Given: Regular Selling Price =_____(1)_____
   Sale Price =_____(2)_____
B. A Cellphone Company buys Vivo phones at PhP8,500. Expenses are set at 18% of the selling Price and the required profit is set at 20% of the selling price. During the weekend sale, the phone is sold at 10% discount. What is the regular selling price of the phone?
   Given: Cost = _____(3)_____
   Rate of Expenses = _____(4)_____
   Rate of Profit = _____(5)_____
   Rate of Discount = _____(6)_____
C. Appliance store realizes a mark-up of P3,250 in selling washing machines. If the items are sold at a mark-up of 40% of the selling price, what is the regular selling price of the washing machine?
   Given: Amount of Mark-up = _____(7)_____
   Rate of Mark-up based on Selling Price=_____(8)_____
D. An entrepreneur bought 225 blouses at PhP15,750. If the blouses were sold at a mark-up of 150% based on cost, what was the amount of mark-up realized on the purchase?
   Given: Cost = _____(9)_____
   Rate of Mark-up based on cost = _____(10)_____

Activity 3. Stepping Stone
Instructions: In each of the real life situations below, list down the steps to answer the question asked.

Problems	**Steps**
A.I am waiting for a pair of sunglasses currenttly priced at PhP 1,200 to be sold at a sale price of P850. *1. What rate of mark down will be*	

Academic Senior High School of the City of Meycauayan
St. Francis (Gasak), City of Meycauayan, Bulacan
Subject Teacher: **Mrs. Lailah Claire U. Ronidel**
Contact Number: **09684694646**
Facebook Messenger Account: **Ma'am Lailah Ronidel**

BUSINESS MATH QUARTER 3

Most Essential Learning Competencies	Output To Be Submitted	Date of Completion
Quarter III Week 1 ***2 Fractions/ decimals and percentage*** **February 21-25, 2022** ✓ Give real-life situations to illustrate fractions, decimals, and percent (ABM_BM11FO-Ic-4) ✓ Solve problems involving fractions, decimals and percent (ABM_BM11FO-Id-5)	**Worksheet for Week 1 -2 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 1-2 . Gauge. Activity 8. Convert Me! (A, B, C, D, E, F) o Write all your answers on yellow paper. o Write the activity title o Write the quarter number and week number on the top right side of the paper.	**March 3, 2022**
Quarter III Week 3 – ***Key Concepts of Ratio and Proportion*** **February 28 – March 4, 2022** ✓ Identify the different kinds of proportions and write examples of real-life situations for each	**Worksheet for Week 3 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 3. Explore. Activity 1. #s 1-5	**March 7, 2022**
Quarter III Week 4 ***– Key Concepts of Ratio and Proportion*** **March 7-11, 2022** ✓ Solve problems involving direct, inverse and partitive proportion (ABM_BM11RP-If-4)	**Worksheet for Week 4 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 4. Gauge #s 1-15	**March 14, 2022**
Quarter III Week 5 ***Key Concepts of Buying and Selling*** **March 14-18, 2022** ✓ Differentiate Mark-on, Mark-down, and Mark-up obtain Mark-on, Mark-down and Mark-up of a product (ABM_BM11BS-Ig-1)	**Worksheet for Week 5 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 5a. Explore. Activity 1. Can You Tell? #s 1-10 Module 5b, Explore. Activity 1. Complete Me! Letters – A, B, C	**March 21, 2022**
Quarter III Week 6 ***Key Concepts of Buying and Selling*** **March 21-25, 2022** ✓ Differentiate Mark-up from margins (ABM_BM11BS-Ih-3) ✓ Describe how gross margins is used in sales (ABM_BM11BS-Ih-4) ✓ Compute single trade discounts and discount series	**Worksheet for Week 6 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities Module 6. Gauge #s 1-10 Module 7. Gauge #s 1-10 Module 8. Enrichment. Activity 1. Solve Me #s 1-3	**March 28, 2022**

Quarter III Week *7 Key Concepts of Buying and Selling* **March 28- April 1, 2022** ✓ Differentiate profit from loss (ABM_BM11BS-Ii-6) ✓ Illustrate how profit is obtained and how to avoid loss in a given transaction (ABM_BM11BS-Ii-7)	**Worksheet for Week 7 (Reference Activity Sheet Quarter III)** Instructions: Answer the following activities Module 9. Explore #s 1-5 Module 10. Gauge #S 1-10	**April 4, 2022**
Quarter III Week *8 Key Concepts of Buying and Selling* **April 4-8, 2022** ✓ Determine the break-even point ✓ Solve problems involving buying and selling products (ABM_BM11BS-Ij-9)	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 11. Explore. Activity 1. Is it a Fact or a Bluff?	**April 11, 2022**
Quarter III Week 9 Key Concepts of Buying and Selling **April 11-13, 2022** ✓ Compute interest specifically as applied to mortgage, amortization and on services/ utilities and on deposits and loans	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 13. Gauge # 1-15	**April 18, 2022**